

Hybrid Renewable Energy System potentials in rural areas: A case of Zambia

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Abstract: The use of conventional system such as hydropower plant is no longer sufficient to meet the increasing demand for electricity in Lusaka, Zambia. Therefore, the present study aims to investigate a hybrid system using the available renewable resources (biomass, wind, and solar) in Misisi rural area. To achieve this, a techno-economic analysis is conducted using HOMER Software. The results obtained showed that the hybrid system produces an output power of 73,577 kWh/yr for an estimated consumption of about 60,386 kWh/yr. Moreover, in the hybrid system, the efficiencies of the biomass, solar PV, and wind generators are 37%, 26.9%, and 36.1% respectively. The estimated total capital cost of the whole system is 151,708.84 USD. Conclusively, the results obtained showed that Zambia has the required renewable energy to produce power that can be used to supplement the conventional systems. These findings also revealed that by increasing the uptake of off-grid projects in rural areas in Africa, could lead to many benefits such as improved education, time saving, and job creation.

Keywords: Solar PV, Wind, Biomass, Energy access, Electricity.

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I. INTRODUCTION

Zambia is one of the developing countries in Africa. The country consists of ten provinces, Central, Copperbelt, Eastern, Luapula, Lusaka, Muchinga, Northern, North-Western, Southern, and Western, as shown in figure 1. Zambia has a population of about 16.2 million people, with a projected increase of more than 25 million by 2030. The population has risen from around ten million in 2010 to sixteen million in 2016 [1]. This as a result has led to a huge increase in energy demand. Approximately 28% of the Zambia population now has access to electricity (62% in urban and 5% in rural areas) [1]. Conversely, 72% of this population does not have access to electricity. The most affected are the people living in rural areas, where candles and light torches are used extensively. About 16% of the Zambians population uses clean cooking fuels and technologies, while the majority use firewood or coal as their primary cooking fuel [2]. In Lusaka, 96% of rural areas use a solar panel for lighting, while some households use torch as their source of lighting. In a previous study, Atteridge et al. [3] interviewed the people in Lusaka about electricity access. In the case of household without a grid connection, a participant in the study explained that although she had the capacity and willingness to pay tariffs if she were to use electricity just for lighting, but she could not afford the connection fee.

Other participants also said that their use of electricity for lighting depend on equipment such as refrigeration and/or television. For cooking, electricity is used only sparingly and then only for quick tasks such as boiling an egg because the tariffs are unaffordable. Zambia has approximately 6,000 MW of untapped hydropower potential, with only approximately 2,255 MW has been developed so far. Furthermore, the available renewable energy resources such as solar, biomass, wind, and geothermal are under-utilised. The aim of the present study is to harness the potential of these energy resources to increase access to electricity in rural areas, especially in the Misisi region. The population of Misisi is estimated to be around 10,000. The study also investigates the techno-economic feasibility of these renewable energies in a combined hybrid system. Based on the energy resources and household energy demand in Misisi rural area, these investigations will be carried out using HOMER Pro software. The work performed in this study is organised as follows: Section 1 covers the introduction. Section 2 presents the renewable energy potential and conversion methods. Section 3 covers the simulation of the hybrid system, while section 4 presents the results and discussion. The conclusion of the study is discussed in the last section.

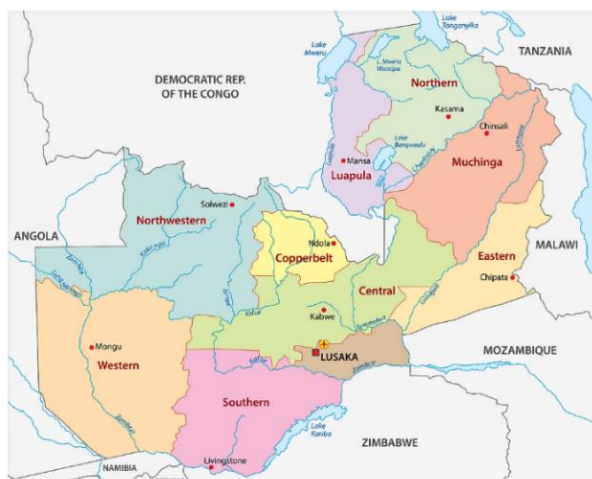


Fig. 1 Zambia Map showing all the provinces.

II. RENEWABLE ENERGY POTENTIAL AND CONVERSIONAL METHODS

A. Biomass Energy

Biomass can be used in solid or gasified form for heating or energy generation. It can also be transformed to liquid or gaseous fuels. Biomass can be used as a source of renewable energy either directly through combustion for the production of heat or indirectly through the transformation of biomass into various forms of biofuels [4]. The biofuel can be used in a gas turbine to generate electricity, a heat-producing boiler, or a combined heat and power plant (CHP) to generate both heat and power. Various methods for converting biomass to biofuel exist, and they are broadly classed as physical-chemical, combustion, thermochemical, and biochemical transformations [5].

Presently, woody biomass and charcoal are projected to provide 95% of the resources used by Zambian households [6]. Even though Zambia's forest landscape is vast (about 66% of the country's total land area), it is vanishing due to rising energy demand. One of the reasons for investing in renewable fuels like biogas and biofuels is the strain on the forest resource. Jatropha, cottonseed, soy seed, and sunflower seeds are all potential biofuel feedstocks in Zambia [6]. According to the Ministry of Energy, there are currently 341,000 biogas digesters in use in the country. Another significant ways to use biomass energy to complement Zambia's energy needs is the use of sugar cane [7][8]. Sugar cane is grown in three provinces and

processed by three companies, with a total annual capacity of 483,000 tonnes of sugar.

B. Solar Energy

Photovoltaics (PV) and solar thermal systems are two ways to generate electricity from solar energy. When the sun's radiant energy hit the solar cell, it causes the electrons to flow around, resulting in electricity. An electric current is started by the action of electrons. The conversion of sunlight to energy occurs invisibly and quickly. There are no moving parts to break down. The concentrated solar power (CSP) utilises solar radiation to generate electricity in a similar way as solar cells, but in a different method. Most solar thermal systems employ a mirrored solar collector to direct sunlight onto a receiver that heats a liquid. In the same way that coal plants operate, the super-heated liquid is utilized to create steam to generate electricity [9].

Zambia has a year-round average of 2,000 to 3,000 hours of sunshine. Global horizontal irradiation is 5.5 kWh/m²/day on average. The areas with the highest Global Horizontal Irradiation (GHI) are the south-west part of Western Province. The south-east part of Luapula Province has an average daily amount of sunshine exceeding 6.3 kWh/km² (yearly total around 2300 kWh/km²) and more [10][7][11].

C. Wind Energy

A wind turbine is a revolving mechanism that converts mechanical energy from the wind's kinetic energy. The propeller-like blades of a turbine are turned by the wind around a rotor, which spins a generator, and generates power [12]. Wind speeds has an average of 2.5 m/s at 10 m above the ground (a speed that is mainly suitable for mechanical applications). The Department of Energy has drawn plans to develop a wind atlas to identify areas where electricity can be generated from wind. Seven areas in Zambia have been identified as viable for off-grid wind power generation [7].

D. Hydro Energy

To generate electricity using hydropower energy, water must be in motion. This is kinetic (moving) energy. When flowing water turns the blades in a turbine, its form is changed to mechanical (machine)

energy. The turbine then turns the generator rotor which converts this mechanical energy into another energy form (electricity). Since water is the initial source of energy, this is called hydroelectric power [13].

Presently, Zambia has an installed large hydro power generation capacity of 2,350 MW. In the future, the projected capacity is about 6,000 MW [14]. The country depends on hydropower for power generation which accounts about 95% of total generation [7][8].

III. SYSTEM SIMULATION USING HOMER PRO

In the present study, the potential renewable energies (biomass, solar PV and wind) described in section II are combined to form a Hybrid Renewable Energy System (HRES).

A. Load Profile

Understanding the load profile and local resource availability of a community was important in getting the right energy solution combination. The load profile is a pattern of energy usage for a consumer or end-user over a given period. The load profile for Lusaka showed that the total load demand obtained is 165 kWh/day. From figure 2, the load demand start increasing from 8 AM and get worse in the evening until around 9 PM daily. Furthermore, it increases more in the evening due to activities such as cooking, lighting usage, entertainment/social (radios and TVs), etc.

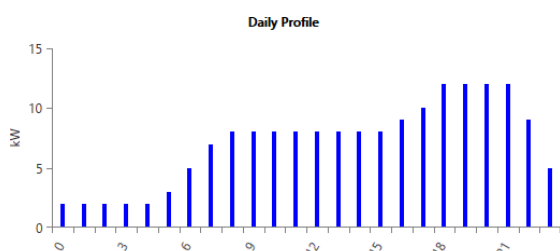


Fig. 2 Load demand graph

B. HOMER software inputs and outputs schematic

To conduct the simulation, the HRES is predefined, which means that energy sources, storage systems, and energy demand must all be chosen. Figure 3 depicts the setup and the output requirements of the HRES in the HOMER software. The simulation

inputs are used by the system to calculate the hourly energy balances over the course of a year. HOMER software is also used to calculate the energy flow from/to each component (generation, storage, or other energy conversion system) per hour. The power demand is also met by the power generated at each hour.

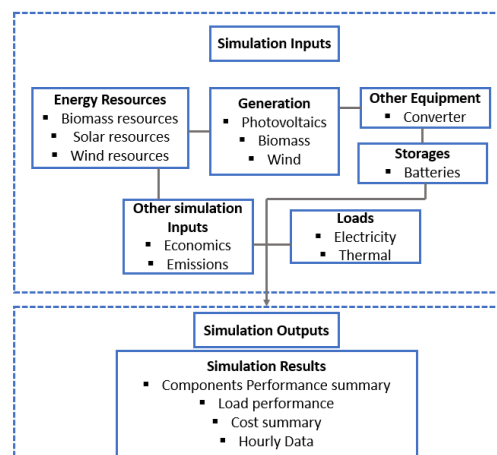


Fig. 3 Schematic diagram showing the input and output parameters defined in HOMER

C. HRES Configuration

The configuration of the HRES consists of three different technologies such as solar PV, wind, and biomass, as shown in figure 4. The mode of operation of this system begins when biogas (from organic waste) was fed into the generator. Subsequently, AC electricity was produced and stored in a 1 kWh LA battery. A 1kW flat plate photovoltaic and a generic 10 kW wind turbine are also coupled to the system to supplement the power produced. The generated electricity in the PV system was DC, while AC electricity is generated from the wind turbine system.

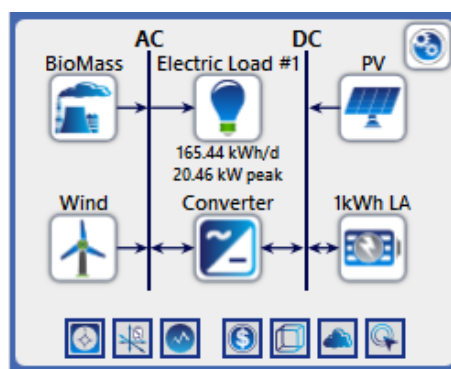


Fig. 4 HRES configuration in HOMER

IV. RESULTS AND DISCUSSION

A. Electric power production

Figure 5 shows the example for a typical week of May when solar PV and wind were not producing much power due to the weather condition. Moreover, biomass gasifier and the power stored in the battery are used to meet the power demand. HOMER also give the power produced and consumed in a year for a place identified in the program which in this study was Misisi in Lusaka

Zambia. Tables 1 and 2 present the electricity produced by the HRES and the load consumed. As seen in table 1, the HRES (solar PV, wind, and biomass generators) produces a total power of 73,577 kWh/yr, with estimated power consumption of 60,386 kWh/yr (Table 2). These results show that the renewable resources in small rural areas like Misisi are enough to generate electricity that could cover the increasing energy demand.

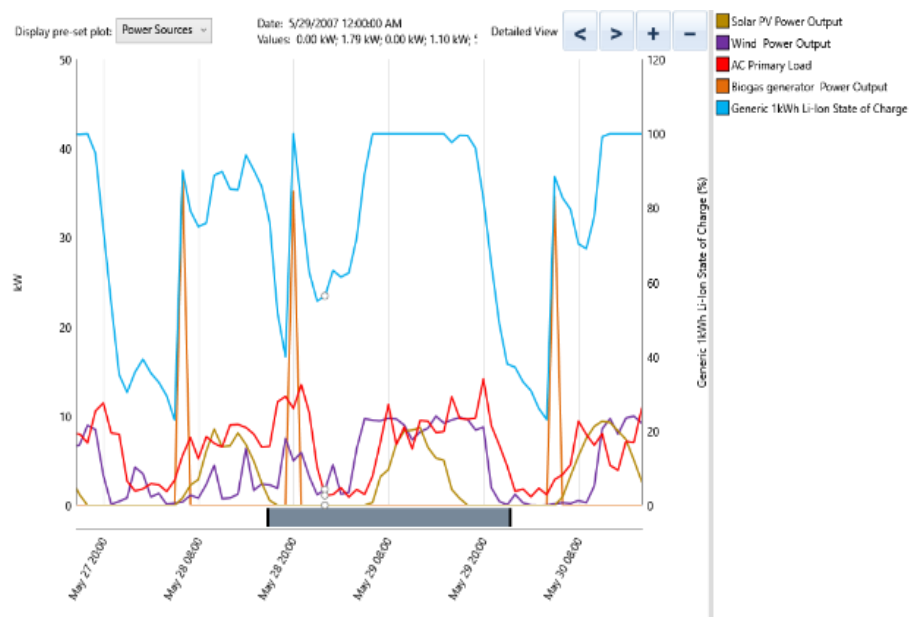


Fig. 5 HOMER hourly operation of the HRES

Table 1 Electric power production

Production	kWh/yr	%
Solar PV	19,812	26.9
Biogas Generator	27,219	37.0
Wind	26,546	36.1
	73,577	100

Table 2 Electric power consumption

Consumption	kWh/yr	%
AC Primary Load	60.386	100
DC Primary Load	0	0
Deferrable Load	0	0
Total	60.386	100

Considering Tables 1 and 2, the output power of 47.031 kWh/yr obtained from the solar PV and biomass generators is comparable to the power (155,693 kWh/yr) reported by [14] [15], where

about 109,058 kWh/yr loads was used in Mumbeki village, Zambia.

B. Component's efficiency

The efficiency of the biomass, solar PV, and wind generators are 37%, 26.9%, and 36.1%, respectively. The solar PV and biomass generator efficiencies are comparable to the efficiency of 15.8% and 23% reported by [14][15]. When comparing these results, the efficiencies obtained is higher in the present study.

C. Cost

A summary of the total net cost in the present system is presented in Table 3. The major cost of the HRES is the capital cost, followed by the replacement cost. The total capital cost is 151,708.84 USD, while the

replacement cost is 78,297.68 USD. The largest capital cost is obtained from the wind system, followed by biomass, solar

PV, battery, and the converter. The yearly operation and maintenance cost is 32,185.19 USD.

Table 3 Components cost

Components	Capital Cost \$	Replacement \$	Operation & Maintenance \$	Total \$
Wind	50,000	4,463.76	6,463.76	63,420.71
Solar PV	28,416.27	0	1,469.41	29,885.68
Biogas generator	40,000	12,118.85	18,563.91	62,984.97
Converter	9,092.56	3,857.74	0	12,224.23
Battery	24,200.00	46,380.72	5,688.11	73,214.00
System	151,708.84	78,297.68	32,185.19	241,729.60

V. CONCLUSION

This paper investigates a hybrid system (HRES) consisting of solar PV, biomass, and wind generators. The study found that Zambia has enough renewable resources to generate electricity for Misisi and other surrounding small communities. The power (73.577 kWh/yr) produced by the HRES was able to meet the load profile peak of 20.46 kW in Misisi region. Moreover, this power can be used to supplement the conventional systems. The outcome of this study also revealed that by increasing the uptake of off-grid projects in Africa, could bring many benefits to the rural and urban areas. Some of these benefits are improved education, time saving, and job creation.

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